

A to Z Preparation Guide for Code with Cisco by Vikram Gaur - LinkedIn Top Voice '24 | Google Cloud Facilitator | SDE @ EY



Securing a 6-month internship + FTE offer at Cisco through the Code-with-Cisco hackathon to hire Software Engineering Summer Interns and Winter Interns (+FTE). Here's a comprehensive guide to help you prepare for the entire process, from the hackathon to the interviews. This guide will help you prepare comprehensively for the different stages of the selection process, including important topics, potential questions, syllabus, and exam structure.

Overview

Code with Cisco is an initiative by Cisco to hire Software Engineering Summer Interns and Winter Interns (+FTE). The hiring process includes a series of assessments and interviews designed to evaluate your technical skills, problem-solving abilities, and cultural fit within the company.

Application Process

- Hackathon Registration: Form a team of 3 members and register for the Code-with-Cisco hackathon.

- Online Assessment: Participate in the online assessment which includes group and individual assessments.
- Interviews: Prepare for multiple interview rounds, including technical, managerial, and HR.

Exam Structure

Online Assessment

- Group Assessment:
 - 3 medium to hard-level coding questions.
 - Duration: 1 hour.
 - Assessment is done as a team.
- Individual Assessment:
 - Multiple-choice questions (MCQs) on CS fundamentals (Networking, OS, Microprocessors).
 - 2 Data Structures and Algorithms (DSA) problems.
 - Duration: Approximately 1-1.5 hours.

Interview Rounds

- Technical Round:
 - Duration: 1 hour 15 mins.
 - Focus: CS fundamentals, practical application of DSA, projects.
- Managerial Round:
 - Duration: 1 hour 20 mins.
 - Focus: 80% technical questions, 20% managerial questions.
- Executive Round:
 - Duration: 1 hour 20 mins.
 - Focus: Deep dive into projects, technical discussions.
- Emerging Talent Round (ETR):
 - Duration: 15 mins.
 - Focus: HR questions, future aspirations.

Syllabus and Important Topics

CS Fundamentals

- Operating Systems (OS):
 - Process and Thread Management.
 - Semaphores and Deadlocks.
 - Memory Allocation (Stack vs. Heap).

- Scheduling Algorithms.
- Computer Networks (CN):
 - OSI Model and TCP/IP Model.
 - DNS, IP Addressing, Subnetting.
 - Network Protocols (TCP vs. UDP).
 - Client-Server Architecture.
- Database Management Systems (DBMS):
 - SQL Queries.
 - Normalization.
 - Transaction Management.

Data Structures and Algorithms (DSA)

- Arrays and Strings.
- Linked Lists (Singly and Doubly).
- Stacks and Queues.
- Trees (Binary Search Tree, AVL Tree).
- Graph Algorithms (DFS, BFS).
- Sorting and Searching Algorithms.

Programming Languages

- Proficiency in at least one programming language (C++, Java, Python).
- Understanding of how to implement data structures and algorithms in the chosen language.

Potential Questions and Preparation Tips

Technical Round Questions

1. Implement Stack using Queues.
2. Practical questions on semaphores in OS.
3. Explain static and dynamic memory allocation.
4. Discuss a favorite project and dive into technical details.

Tip: Be familiar with Cisco products and prepare questions to ask the interviewer.

Managerial Round Questions

1. Medium-level problem on doubly linked list.
2. Discussion on why Linux is preferred over Windows.
3. In-depth questions about classful addressing, multicasting, IP address.
4. Explain the process when you type "cisco.com" in a web browser.

Tip: Be thorough with your projects and be ready to discuss alternatives and technical challenges faced.

Executive Round Questions

1. Implement a Python library from scratch based on your project.
2. Deep dive into learning approaches and technical blogs.

Tip: Lead the discussion to highlight your strengths and interesting projects.

HR Round Questions

1. Why Cisco?
2. Where do you see yourself in 2 years?
3. Discuss Cisco competitors.

Tip: Prepare answers to common HR questions and understand Cisco's work culture.

Resources for Preparation

- Coding Practice: Leetcode, GeeksforGeeks.
- OS and CN: Operating Systems by William Stallings, Computer Networking by James F. Kurose and Keith W. Ross.
- DBMS: Database System Concepts by Silberschatz, Korth.
- Aptitude: Practice basic quantitative and logical reasoning problems.
- Interview Prep: Mock interviews, discussion with peers, and online interview experiences.

1. Hackathon and Initial Screening

Group Assessment

- Structure: Teams of three participants.
- Content: Medium to hard-level coding questions.
- Time: 1 hour.
- Goal: Solve as many problems as possible within the time limit. Typically, solving all problems correctly will give you a strong chance to proceed to the individual assessment phase.

Tips:

- Practice coding challenges on platforms like LeetCode, HackerRank, and Codeforces.
- Focus on medium to hard-level problems.
- Work on team communication and collaboration skills, as the group assessment tests how well you work in a team.

2. Individual Assessment

Content:

- MCQs: Covering CS fundamentals such as Networking, Operating Systems, Microprocessors, etc.
- Coding Questions: Typically 2 DSA-based problems.

Topics to Cover:

- Data Structures: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Heaps, Hash Tables.
- Algorithms: Sorting (Quick, Merge, Heap), Searching (Binary Search), Dynamic Programming, Greedy Algorithms, Backtracking.
- Networking: OSI Model, TCP/IP, DNS, IP addressing, Subnetting.
- Operating Systems: Processes and Threads, Scheduling Algorithms, Memory Management, Deadlocks, Semaphores.
- Microprocessors: Basic architecture, instruction sets, and functioning.

3. Interview Rounds

Technical Round

- Duration: 1 hour 15 minutes.
- Focus Areas:
 - CS fundamentals (OS, CN, DBMS, DSA).
 - Practical application-based questions.
 - In-depth discussion of your projects.

Sample Questions:

- Implement a stack using queues.
- Practical questions on semaphores.
- Explain static and dynamic memory allocation.
- Discuss your favorite project and dive into technical details.

Tips:

- Be thorough with your basics.
- Be ready to discuss your projects in detail.
- Prepare questions about Cisco's products to ask at the end of the interview.

Managerial Round

- Duration: 1 hour 20 minutes.
- Focus Areas:
 - Combination of technical questions and managerial scenarios.
 - Deep discussions on your projects.

Sample Questions:

- Medium-level doubly-linked list problem.
- Security advantages of Linux over Windows.
- Detailed discussions on classful addressing, multicasting, IP addresses.
- Troubleshooting network issues.

Tips:

- Prepare answers for common managerial questions.
- Be prepared to discuss your problem-solving and leadership experiences.

Executive Round

- Duration: 1 hour 20 minutes.
- Focus Areas:
 - Implementation details of technologies used in your projects.
 - Approach to learning new topics.

Sample Questions:

- How would you build a Python library from scratch?
- Deep dive into technical projects mentioned in your resume.

Tips:

- Lead the discussion towards your strengths.
- Be prepared to discuss how you approach learning and problem-solving.

ETR Round (Emerging Talent Round)

- Duration: 15 minutes.
- Focus Areas:
 - Motivation for joining Cisco.
 - Career aspirations.

Sample Questions:

- Why Cisco?
- How do you see yourself in 2 years?
- Plans for pursuing further education.
- Discussion on competitors and industry landscape.

Tips:

- Prepare answers to common HR questions.
- Be honest and confident.

4. Resources for Preparation

Coding Practice:

- LeetCode: Focus on medium to hard-level problems.
- HackerRank: Practice coding challenges and algorithms.

- GeeksforGeeks: Read through coding interview experiences and technical articles.

CS Fundamentals:

- Operating Systems: "Operating System Concepts" by Silberschatz, Galvin.
- Computer Networks: "Computer Networking: A Top-Down Approach" by Kurose and Ross.
- DBMS: "Database System Concepts" by Silberschatz, Korth.
- Microprocessors: "The 8051 Microcontroller and Embedded Systems" by Mazidi.

Project Preparation:

- Be prepared to discuss all aspects of your projects.
- Understand the technologies used and be ready to discuss alternatives and implementation details.
- Be able to explain the challenges faced and how you overcame them.

5. Final Tips

- Only include projects and skills on your resume that you are confident about.
- Practice coding problems regularly.
- Stay updated on Cisco's latest products and technologies.
- Maintain a positive attitude and be confident during the interviews.

Good luck with your preparation!

Conclusion

Preparing for the Code-with-Cisco program requires a thorough understanding of computer science fundamentals, hands-on coding practice, and good communication skills. Make sure to review your projects comprehensively and practice commonly asked interview questions.

Good luck with your preparation!

Cisco Interview Experience G4 (Code with Cisco)

Online Assessment Questions:

1. Coding Questions:

- First problem: Similar to <https://codeforces.com/problemset/problem/1132/F>
- Second problem: Similar to <https://cses.fi/problemset/task/1097>.
- Third problem: A simple greedy algorithm problem

Onsite Hackathon:

1. Problem Statements:

The hackathon had three problem statements –

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1. The first statement asked participants to make a visualisation tool for a given graph.
2. The second statement asked participants to make a game related to Cyber Security.
3. The last statement was related to the Dynamic Connectivity Problem of Graphs.

Each problem statement had a winning team, who won a Go-Pro for each team member.

Interview Questions:

Round 1 – Technical:

1. C Language:
 - Function pointers
 - Pointers
 - Memory allocation
 - Dynamic memory
 - Memory leaks
2. Operating Systems:
 - Booting process
3. Database Management System:
 - Basic DBMS concepts
4. Computer Networks:
 - Difference between IP address and MAC address (NIC Card details)
5. Data Structures:
 - Stack
 - Queue
6. Python:
 - Questions related to Python as mentioned in the resume

Round 2 – Technical + HR:

1. Detailed discussion about internships and projects listed on the resume.

Round 3 – Executive Round (Two Interviewers):

1. Object-Oriented Programming Systems (OOPS):
 - Basic OOPS concepts
2. Projects:
 - Discussion on the projects listed on the resume
3. Sorting:
 - Sorting an array in $O(n)$ time using a different data structure (Inorder traversal of a Binary Search Tree)
4. Coding Question:

- Intersection of two linked lists (Initially given the wrong problem, then fixed and discussed the solution)

5. General Questions:

- Why do you want to join Cisco?

Round 4 – ETR Connect:

1. Relocation:

- Willingness to relocate to Bangalore (BLR)

2. Availability:

- Availability for a 6-month internship starting in January

3. Stipend and Salary:

- Discussion on stipend and salary

Resources for Preparation:

Coding Practice:

- *Codeforces*: Practice problems similar to the ones mentioned in the online assessment.
- *CSES Problem Set*: Particularly useful for dynamic programming problems.
- *LeetCode*: For practicing intersection of linked lists and other common coding problems.
- *GeeksforGeeks*: For a variety of coding and technical interview questions.
- *InterviewBit*: For practicing coding and data structure questions.

C Language:

- "The C Programming Language" by Brian W. Kernighan and Dennis M. Ritchie
- GeeksforGeeks C tutorials

Operating Systems:

- "Operating System Concepts" by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne
- GeeksforGeeks OS tutorials

Database Management Systems (DBMS):

- "Database System Concepts" by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan
- GeeksforGeeks DBMS tutorials

Computer Networks:

- "Computer Networking: A Top-Down Approach" by James F. Kurose and Keith W. Ross
- GeeksforGeeks CN tutorials

Python:

- "Python Crash Course" by Eric Matthes
- GeeksforGeeks Python tutorials

Data Structures and Algorithms:

- "Introduction to Algorithms" by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
- GeeksforGeeks DSA tutorials

Object-Oriented Programming:

- "Object-Oriented Design & Patterns" by Cay S. Horstmann
- GeeksforGeeks OOPS tutorials

General Interview Preparation:

- "Cracking the Coding Interview" by Gayle Laakmann McDowell
- Mock interviews on platforms like Pramp or Interviewing.io

Using these resources and understanding the types of questions asked can significantly enhance your preparation for a Cisco interview or any similar technical interview.

Cisco Interview Experience for Software Engineer

Round 1: Online Test (70 minutes)

- Platform: HackerRank
- Format:
 - 15 MCQs: Topics included Electronics, Computer Networks, Aptitude, and Data Structures (No negative marks).

- 2 Coding Questions:

1. Longest non-repeating substring -

<https://www.geeksforgeeks.org/length-of-the-longest-substring-without-repeating-characters/>

2. The expression parenthesized in such a way that it gives max value for the expression. Expression has +, -, *

Round 2: Technical Interview

- Roles Offered: CX-Platform, CX-PSS, IT, Engineer, and SCO (Selected for IT).
- Format:

- Panel Interview: 3 panelists (P1, P2, P3)

- Questions:
 1. Introduce yourself.
 2. Explain your project.
 3. Difference between Binary tree and Trie data structures.
 4. Logical question: Using the digit 8 (as 88888888), use only the + operator to sum to 1000.
 5. Data Structures proficiency (rate yourself).
 6. Code for:
 - Finding the middle node of a linked list.
 - Given an array of size [1...1000], find 2 missing numbers.
 7. Simple database queries.

Round 3: Managerial Interview

- Format:
 - Introduce yourself.
 - Feedback from technical interviewers.
 - Brief on projects.
 - Questions about hobbies, college campus, why Cisco, and career aspirations.
 - Questions to the interviewer:
 1. What exactly is the IT profile in Cisco?
 2. How is the culture in Cisco?
 3. What is the trend of IT in the future?

Round 4: HR Interview

- Format:
 - Internship availability in January.
 - Comfort with working 5 days a week.
 - Plans for higher studies.
 - Comfort with relocation to Bangalore.

Result: PLACED at CISCO

Preparation Strategy

- Learning C++: Started in the 5th semester.
- Basic Algorithms: Learned from GeeksforGeeks.

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- Practice Platforms: HackerRank, LeetCode, GeeksforGeeks.
- Focused Topics: Linked lists, trees.
- Resources: Videos of Tushar Roy.

Sample Interview Questions and Resources

1. Longest Non-Repeating Substring:

- [GeeksforGeeks - Longest Substring Without Repeating Characters](https://www.geeksforgeeks.org/length-of-the-longest-substring-without-repeating-characters/)
- [LeetCode - Longest Substring Without Repeating Characters](https://leetcode.com/problems/longest-substring-without-repeating-characters/)

2. Expression Parenthesization (Matrix Chain Multiplication):

- [GeeksforGeeks - Matrix Chain Multiplication](https://www.geeksforgeeks.org/matrix-chain-multiplication-dp-8/)

3. Find Middle of Linked List:

- [GeeksforGeeks - Find the Middle of a Given Linked List](https://www.geeksforgeeks.org/find-the-middle-of-a-given-linked-list/)

4. Find Missing Numbers in an Array:

- [GeeksforGeeks - Find Two Missing Numbers](https://www.geeksforgeeks.org/find-two-missing-numbers-set-1-an-interesting-linear-time-solution/)

5. Database Queries:

- [GeeksforGeeks - SQL Interview Questions](https://www.geeksforgeeks.org/sql-interview-questions/)

General Tips for Cisco Interviews

- Resume:
 - Keep it concise and relevant.
 - Highlight key projects and skills.
- Practice Regularly:
 - Use coding platforms to solve a variety of problems.
- Understand Fundamentals:
 - Deepen knowledge in data structures, algorithms, OS, DBMS, and networks.
- Mock Interviews:

- Simulate the interview environment for better preparation.
- During the Interview:
 - Clearly communicate your thought process.
 - Stay calm and composed; admit if you don't know an answer.

Common Mistakes to Avoid

1. Inconsistent Practice:
 - Regular practice is essential for improvement.
2. Ignoring Fundamentals:
 - Ensure a strong grasp of basic concepts.
3. Lack of Behavioral Preparation:
 - Prepare for HR and behavioral questions as well.
4. Not Asking Questions:
 - Prepare thoughtful questions for the interviewers.

By covering all these aspects, you should have a comprehensive guide to preparing for a Cisco interview. Good luck!

General Advice for Interviews:

- Resume Tips:

- Keep it concise, usually one page.
- Highlight relevant projects, internships, and skills.
- Use bullet points for clarity and impact.

- Preparation Strategy:

- Practice coding regularly on platforms like LeetCode, Codeforces, and HackerRank.
- Review basic and advanced data structures and algorithms.
- Understand the fundamentals of operating systems, databases, and computer networks.
- Mock interviews can help simulate the interview environment and improve confidence.

- During the Interview:

- Communicate your thought process clearly.
- Ask clarifying questions if the problem statement is not clear.
- Take a structured approach to solve problems.
- Don't be afraid to make educated guesses if you're unsure about an answer, but explain your reasoning.

Post-Interview:

- Follow-up:
 - Send a thank-you email to the interviewers, expressing gratitude for the opportunity and reaffirming your interest in the position.

- Reflect:

- Analyze your performance in the interview and identify areas for improvement.
- Continue practicing and learning, regardless of the outcome.

Common Mistakes to Avoid:

1. Not practicing enough coding problems:
 - Consistent practice helps in improving problem-solving speed and accuracy.
2. Ignoring fundamental concepts:
 - A solid understanding of basic concepts is crucial for tackling complex problems.

3. Lack of preparation for behavioral questions:

- Technical interviews also include HR and behavioral questions to assess cultural fit.

4. Not asking questions:

- Always prepare a few questions to ask the interviewer about the role, team, or company.

I've dedicated a lot of time and effort to preparing a comprehensive guide on how to ace the Cisco Software Engineer interview & Code with Cisco. From detailed interview experiences to essential preparation tips and resources, I've compiled everything you need to succeed.

If you find my guide helpful, please consider buying me a coffee. Your support will enable me to continue creating helpful guides and sharing valuable experiences with you.

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Interview prep resources

OS RESOURCES		
Codehelp's playlist	Operating Systems for Placements 2022 - YouTube	for interview centric prep
Gatesmashers's playlist	Operating System (Complete Playlist) - YouTube	use this if studying os for the first time, this will help in interview prep as well
Codehelp notes	■ OS_Full_Notes.pdf	
OS Bedtime stories	■ Bedtime Stories on Operating Systems.pdf	
OS interviewbit	40+ Operating System Interview Questions (2023) - Interviewbit	

CN Resources		
Gate smasher's playlist	https://www.youtube.com/playlist?list=PLxCzCOWd7aiGFBD2-2joCpWOLUrDLvVV	
CN Imp interview topics	■ cn	
Interviewbit CN interview questions	https://www.interviewbit.com/networking-interview-questions/	

DBMS RESOURCES		
codehelp's interview centric playlist	DBMS Placements Series 2022 - YouTube	use this when preparing for interviews
codehelp notes	■ DBMS_Full_Notes.pdf	
gate smashers college exam centric playlist	DBMS (Database Management system) Complete Playlist - YouTube	use this when studying dbms for the first time, this will help in interview prep as well
interviewbit interview questions	https://www.interviewbit.com/dbms-interview-questions/	

SQL RESOURCES	
Leetcode study plan of 50 questions	SQL 50 - Study Plan - LeetCode
Codehelp SQL video	■ Lecture 9: Complete SQL in 1 VIDEO - YouTube

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